



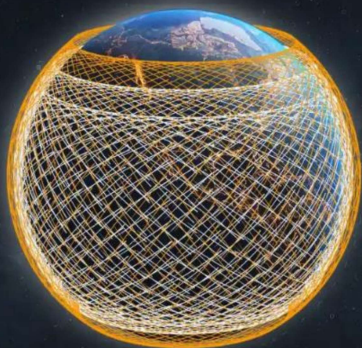
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Amazon Leo Satellite System

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Overview of Amazon Leo Mega Constellation



Amazon Leo Constellation showing 3 Altitude Shells and Inter-satellite Links

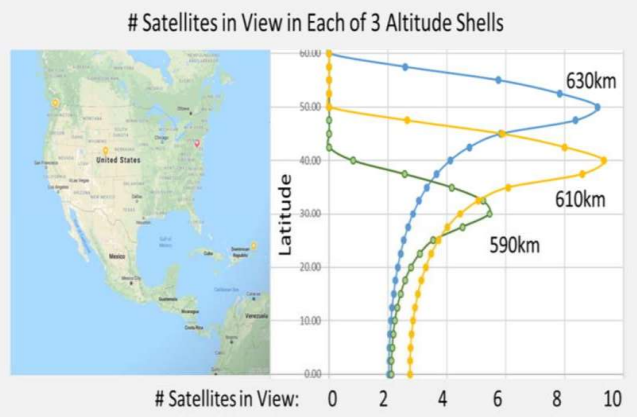
Constellation Layout

Altitude Shell (km)	Inclination (deg)	Orbital Planes	Numbers of Sats
630	51.9	34	1156
610	42	36	1296
590	33	28	784

Expected Completion by July 2029



Satellite Honeycomb Field-of-view (FOV)



Orbital Shells Service Different Latitudes

- **Purpose:** Deliver broadband internet connectivity to users worldwide
- Designed to be launch platform agnostic
 - 10 billion USD committed to launch contracts for constellation expansion; ~45% is Blue Origin
 - 210+ production sats in orbit as of early 2026
- Expected to take between 10 and 15 kW to power; batteries charged via solar cells
- Satellites equipped with phased arrays, dish and patch antennas, and IR lasers for links
- At least 300 gateway stations are expected
- 3 customer terminal variants are supported:

Antenna Size	Upload / Download Speeds (Mbps)	Power Consumption (W)
7" x 7"	40/100	60
11" x 11"	100/400	90
30" x 20"	400/1000	225

The Antenna Toolkit: Three Links, Three Technologies



Phased Array Antennas used on customer (left) and satellite (right) terminals



Parabolic Reflector Antenna used for gateway link (photo not from Amazon Leo system)



Patch Antenna for TT&C link (photo not from Amazon Leo system)



Artist rendering of Amazon Leo satellite system

	User Link	Gateway Link	TT&C Link
Antenna Type	Phased array (satellite and customer terminal)	Parabolic reflector (ground only)	Patch antennas (satellite surfaces) + dish (ground station)
Steering Method	Electronic (beamsteering)	Mechanical (pre-pointed)	Fixed, multi-surface
Dish/Aperture	3 sizes: 7 in ² , 11 in ² , 19"x30"	1 to 2.4 m	2 to 3 m (ground)
Coverage per satellite	~ 1,560,000 km ²	~3,200,000 km ²	Altitude independent
Max Steer Angle (from nadir)	48.2 degrees	58.8 degrees	65 degrees
Min Elevation Angle	35 degrees	20 degrees	5 degrees
Polarization	RHCP/LHCP	RHCP/LHCP	RHCP/LHCP
Purpose	Broadband service to end users	Backbone backhaul to internet	Telemetry, Tracking and Command

Worst-case Amazon Leo Link Analysis Yields Functional C/N₀ with Margin for Weather

Uplink

Downlink

User Terminal to Sat	Notes	Sat Gateway to Terrestrial	Notes
Frequency (GHz)	28.35 28.35 to 30 GHz	Frequency (GHz)	17.70 17.7 to 20.2 GHz
EIRP (dBW)	48.05 Smallest user terminal at 50% PAE, 60% aperture efficiency	EIRP (dBW)	48.47 Pt from Published PFD [12], 0.45m Dish at 60% Efficiency
G/T (dB/K)	13.01 40 dB Antenna Gain; 500 K System Temp	G/T (dB/K)	22.17 43.1 dB Antenna Gain; 125 K System Temp
Lp (dB)	-182.31 1098 km total path length, 35deg EL	Lp (dB)	-182.71 1842 km total path length, 20deg EL
Lothr (dB)	-14.50 Driven by pointing loss approx	Lothr (dB)	-14.50 Driven by pointing loss approx
k*Bn (dBW/K)	-145.59 200 MHz Bn	k*Bn (dBW/K)	-141.61 500 MHz Bn
Uplink C/N (dB)	9.84	Uplink C/N (dB)	15.04
Overall C/N (dB)		8.70	

Uplink

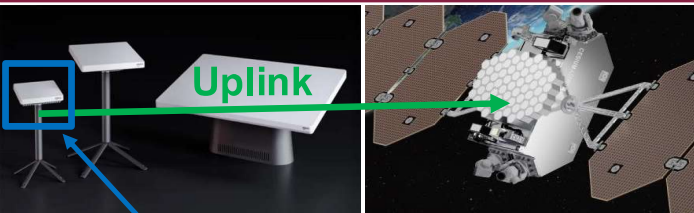
Downlink

Terrestrial Gateway to Sat	Notes	Sat Phased Array to User	Notes
Frequency (GHz)	27.50 27.5 to 30 GHz	Frequency (GHz)	17.70 17.7 to 20.2 GHz
EIRP (dBW)	63.97 Tx power from Starlink [27]; 1m Dish with 60% efficiency	EIRP (dBW)	58.47 Pt from Published PFD [12], 14" square Phased Array
G/T (dB/K)	13.04 0.45m Dish 60% efficiency, 500 K System Temp	G/T (dB/K)	6.88 Smallest User Terminal; 170 K System Temp
Lp (dB)	-186.53 1842 km total path length, 20deg EL	Lp (dB)	-178.22 1098 km total path length, 35deg EL
Lothr (dB)	-14.50 Driven by pointing loss approx	Lothr (dB)	-14.50 Driven by pointing loss approx
k*Bn (dBW/K)	-145.59 200 MHz Bn	k*Bn (dBW/K)	-141.61 500 MHz Bn
Uplink C/N (dB)	21.57	Uplink C/N (dB)	14.25
Overall C/N (dB)		13.51	

Uplink

Downlink

Terrestrial TTC&M to Sat	Notes	Sat TTC&M to Terrestrial	Notes
Frequency (GHz)	27.5 27.5 to 28.05 GHz	Frequency (GHz)	19.25 19.25 to 19.4 GHz
EIRP (dBW)	69.99 Tx power from Starlink [27]; 2m Dish with 60% efficiency	EIRP (dBW)	39.22 Pt from Published PFD [12], 9 dB Gain Patch Antenna
G/T (dB/K)	-17.99 9 dB Antenna Gain; 500 K System Temp	G/T (dB/K)	27.59 49.9 dB Antenna Gain; 170 K System Temp
Lp (dB)	-198.41 7228 km total path length	Lp (dB)	-195.31 7228 km total path length
Lothr (dB)	-3.91 Driven by atmospheric and hardware losses	Lothr (dB)	-3.91 Driven by atmospheric and hardware losses
k*Bn (dBW/K)	-158.60 10 MHz Bn	k*Bn (dBW/K)	-151.61 50 MHz Bn
Uplink C/N (dB)	8.28	Uplink C/N (dB)	19.19
Overall C/N (dB)		7.94	



Array Used in Link Budget Calcs

*For illustrative purposes, this is not an Amazon Leo Sat



0.45m onboard dish
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1-2.4m Dishes, 1m assumed above



Array Used in Link Budget Calcs

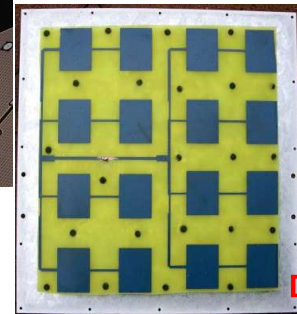
*For illustrative purposes, this is not an Amazon Leo Sat



0.45m onboard dish



1-2.4m Dishes, 1m assumed above



Low Gain, Broad Beam TTC&M Patch Antennas



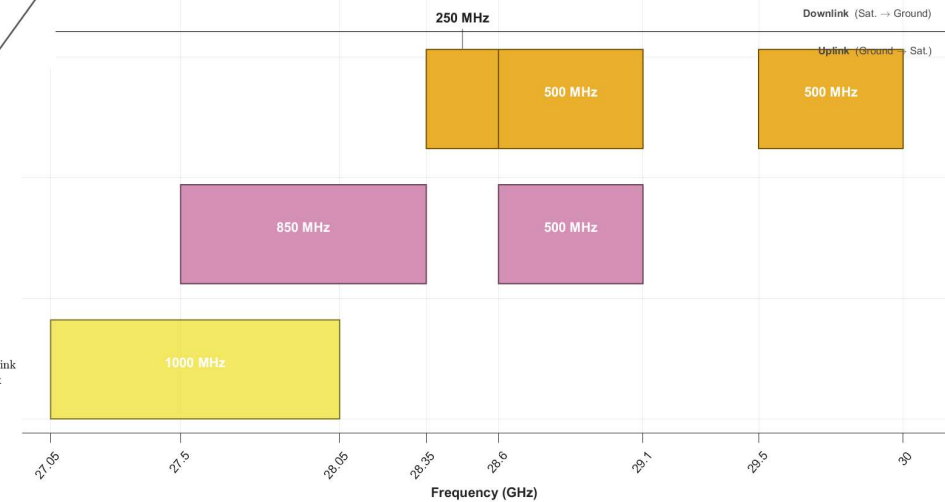
TTC&M Ground Station Dish – 2m Diameter used in Link Budget

Spectrum by the Numbers: Frequency Plan

Kuiper Systems Ka-Band Frequency and Polarization Plan
FCC SAT-LOA-20190704-00057 | Tables 3-5 & Annex D2



- Frequency reuse between User and Gateway downlinks is made possible by SDN beam controller
- Multiple triggers for beam reassignment
 - GSO Exclusion zone
 - Inter-system interference (e.g. Starlink)
 - Satellite handover during orbital motion
- Subchannelization
 - User Downlink (100MHz) – Spatial Frequency reuse, throughput flexibility
 - User Uplink (50MHz) - FDMA/TDMA for frequency slot assignment and time slot scheduling within each slot
 - TT&C (1,5,10,20 or 50MHz) - reduce risk of interference with adjacent allocations



Conclusion

- Amazon's satellites have higher altitudes than Starlink (590 km to 630km vs 340 km to 550 km) and FCC manages frequency interference.
- Amazon to acquire Globalstar (April 2026) to expand their network and use its current position in mobile satellite spectrum (MSS) to break into the Direct-to-Device (D2D) market.
- Leverage AWS for existing secure data transmission network that interconnects consumers globally (9 Mkm of fiber).
- Future use of V-band (42 and 42.5 GHz) to support 5G communication systems, and data throughput from 1Gbps up to 10Gbps